

EPIC 1 — User, Identity & Access Platform

FEATURE 1: User Registration & Login

Description

This feature enables users to create and access an Eduyata account securely.

It is the first control gate of the platform and mandatory for all personas.

It supports self-registration and invite-based onboarding.

It validates credentials and establishes an authenticated session.

It does not assign roles or permissions.

User Journey

1. User lands on the Eduyata login page.
2. User selects register or login.
3. User enters credentials or accepts an invite.
4. System validates credentials and identity.
5. Authentication succeeds or fails.
6. On success, user proceeds to organization and role resolution.

Exit state: Authenticated user with no active access scope.

User Map

Platform Admin uses this feature to access the platform and assist in resolving login failures.

Organization Admin uses it to log in and onboard users. Instructors and learners use it to register and access the platform. Support users use it only to assist with login-related issues and troubleshooting.

FEATURE 2: Single Sign-On (SSO)

Description

This feature allows enterprise users to authenticate using their organization's identity provider.

It removes the need for separate Eduyata credentials and centralizes authentication under the organization.

It does not manage permissions or store passwords.

User Journey

1. User selects organization login.
2. User is redirected to the organization identity provider.
3. Identity provider authenticates the user.
4. Eduyata validates the received token.
5. User identity is resolved or created.
6. User proceeds to role and organization selection.

Exit state: Authenticated via trusted external identity.

User Map

Platform Admin configures and manages Single Sign-On integrations. Organization Admin uses Single Sign-On for their users. Instructors and learners authenticate using Single Sign-On as their primary login method. Support users assist only in troubleshooting authentication failures.

FEATURE 3: Multi-Role Onboarding

Description

This feature supports users who hold multiple roles across one or more organizations.

It ensures a single identity can operate in different capacities without duplication.

It does not grant permissions or override access policies.

User Journey

1. User completes authentication.
2. System detects multiple roles or organizations.
3. User selects the desired role and organization.

4. Role-specific onboarding is displayed.
5. Selected role becomes active.

Exit state: User enters the platform in a clear role context.

User Map

Platform Admin has full visibility into multi-role assignments. Organization Admin assigns and manages roles within their organization. Instructors may select between roles if permitted. Learners typically operate with a single role. Support users can view role context for troubleshooting but cannot change it.

FEATURE 4: Role-Based Access Control (RBAC)

Description

RBAC defines what actions users can perform based on assigned roles.

It enforces access consistently across the user interface, APIs, and data layer.

It does not authenticate users or manage sessions.

User Journey

1. User role is resolved.
2. RBAC engine evaluates permissions.
3. Authorized actions are allowed.
4. Unauthorized actions are blocked.
5. User interface adapts dynamically.

Exit state: Secure, permission-controlled access.

User Map

Platform Admin defines and manages roles and permissions globally. Organization Admin manages role permissions within their tenant scope. Instructors and learners consume permissions but cannot modify them. Support users have no role control authority.

FEATURE 5: User Profile Management

Description

This feature allows users to manage personal profile details while enabling administrators to manage organization-specific attributes.

It separates personal identity from access control.

User Journey

1. User opens profile settings.
2. User edits permitted fields.
3. System validates changes.
4. Updates are saved successfully.

Exit state: Profile information updated.

User Map

Platform Admin can view and edit all user profiles. Organization Admin can manage organization-related profile attributes. Instructors and learners can update their own personal details. Support users have read-only access for troubleshooting.

FEATURE 6: Organization & Tenant Management

Description

This feature enables Eduyata to support multiple independent organizations with strict data isolation.

It is foundational for enterprise and institutional usage.

User Journey

1. Platform Admin creates a tenant.
2. Organization Admin configures organization settings.
3. Users are assigned to the organization.
4. Tenant isolation rules are enforced.

Exit state: Fully isolated organizational environment.

User Map

Platform Admin has full control over tenant creation and configuration. Organization Admin manages users and settings within their tenant. Instructors and learners do not access tenant management. Support users can view tenant data for operational purposes.

FEATURE 7: User Lifecycle Management

Description

This feature controls the user lifecycle from invitation to deactivation.

It ensures access is revoked immediately when status changes.

User Journey

1. User is invited.
2. User activates account.
3. User may be suspended or deactivated.
4. System revokes access instantly.

Exit state: Correct lifecycle state enforced.

User Map

Platform Admin manages lifecycle events globally. Organization Admin manages lifecycle states within their organization. Instructors and learners do not control lifecycle actions. Support users assist only under admin direction.

FEATURE 8: Session & Device Management

Description

This feature tracks active sessions and devices and enforces security policies.

User Journey

1. User logs in.
2. Session is created and device is recorded.
3. Policies are enforced.
4. Session may be revoked if required.

Exit state: Secure active session.

User Map

Platform Admin can view and revoke all sessions. Organization Admin can view sessions within their tenant. Instructors and learners manage their own sessions. Support users can view sessions for investigation.

FEATURE 9: Account Recovery & Credential Management

Description

This feature enables secure account recovery without compromising security.

User Journey

1. User requests recovery.
2. Identity verification occurs.
3. Credentials are reset.
4. User regains access.

Exit state: Account access restored.

User Map

Platform Admin oversees recovery mechanisms. Organization Admin assists users when needed. Instructors and learners initiate self-recovery. Support users assist under defined workflows.

FEATURE 10: Identity Federation Across Products

Description

Allows a single identity to access multiple Eduyata products seamlessly.

User Journey

1. User logs into one product.
2. Identity token is reused.
3. Permissions are revalidated per product.

Exit state: Seamless cross-product access.

User Map

Platform Admin configures identity federation. Instructors and learners use federated identity transparently. Organization Admin and support users do not manage federation directly.

FEATURE 11: Access Scope & Context Switching

Description

Allows users to switch roles, organizations, or products without logging out.

User Journey

1. User initiates context switch.
2. Session scope is recalculated.
3. Permissions are refreshed.

Exit state: New access context active.

User Map

Platform Admin, Organization Admin, and Instructors can switch access scope when permitted. Learners typically do not have context switching privileges. Support users switch context for investigation.

FEATURE 12: Impersonation & Delegated Access

Description

Allows authorized users to temporarily access another user's account view for support and administration.

All actions are audited and time-bound.

User Journey

1. Authorized user requests impersonation.
2. System validates permission.

3. Temporary session is created.
4. Session expires automatically.

Exit state: Impersonation completed and logged.

User Map

Platform Admin and Support users can impersonate other users under strict controls.

Organization Admin, Instructors, and Learners cannot impersonate any user.

EPIC 2 — Security, Privacy & Compliance

FEATURE 1: Multi-Factor Authentication (MFA / Two-Factor Authentication)

Description

This feature adds an additional verification step during login to reduce the risk of unauthorized access.

It strengthens account security beyond passwords by requiring something the user has or verifies.

It applies selectively based on policy, role, or risk level.

It does not replace primary authentication.

User Journey

1. User enters valid login credentials.
2. System checks MFA requirement policy.
3. User is prompted for second factor.
4. User completes verification.
5. Access is granted or denied.

Exit state: Authenticated with strengthened assurance.

User Map

Platform Admin defines MFA policies globally. Organization Admin enforces MFA for their users. Instructors and learners complete MFA during login when required. Support users assist with MFA recovery under strict controls.

FEATURE 2: Risk-Based Login & Anomaly Detection

Description

This feature evaluates login behavior in real time to detect unusual or risky access attempts. It dynamically adjusts security requirements based on context.

It does not block users arbitrarily.
It responds only to calculated risk.

User Journey

1. User attempts login.
2. System evaluates device, location, and behavior.
3. Risk score is calculated.
4. Additional verification or denial is triggered if needed.
5. Login proceeds or is blocked.

Exit state: Risk-adjusted authentication outcome.

User Map

Platform Admin defines risk rules and thresholds. Organization Admin views risk alerts for their users. Instructors and learners experience additional verification when risk is high. Support users review anomalies during investigations.

FEATURE 3: Account Lockout & Brute-Force Protection

Description

This feature prevents repeated unauthorized login attempts by temporarily locking accounts after defined failures.

It protects credentials from automated and manual attacks.

User Journey

1. User submits incorrect credentials repeatedly.
2. System tracks failed attempts.
3. Lockout threshold is reached.
4. Account is temporarily locked.
5. User follows recovery flow to regain access.

Exit state: Account protected from attack.

User Map

Platform Admin configures lockout policies. Organization Admin monitors locked accounts. Instructors and learners may experience temporary lockout. Support users assist with account unlocks.

FEATURE 4: CAPTCHA & Bot Protection

Description

This feature prevents automated bots from abusing authentication and public-facing endpoints.

It protects platform availability and integrity.

User Journey

1. User performs a sensitive action.
2. System detects bot-like behavior.
3. CAPTCHA challenge is triggered.
4. User completes challenge or is blocked.

Exit state: Human interaction verified.

User Map

Platform Admin configures CAPTCHA rules. Organization Admin is not involved. Instructors and learners may encounter CAPTCHA during suspicious activity. Support users troubleshoot false positives.

FEATURE 5: Fraud & Abuse Detection Engine

Description

This feature detects misuse patterns such as fake accounts, credential sharing, and policy abuse.

It operates continuously and silently.

User Journey

1. User activity is monitored.
2. Behavioral patterns are analyzed.
3. Abuse signals are detected.
4. Alerts or restrictions are applied.

Exit state: Platform abuse contained.

User Map

Platform Admin monitors fraud signals globally. Organization Admin receives alerts for their users. Instructors and learners may face restrictions if abuse is detected. Support users assist investigations.

FEATURE 6: KYC / User Verification Framework

Description

This feature verifies user identity using formal documentation when required by policy or regulation.

It is applied selectively, not universally.

User Journey

1. User is prompted for verification.
2. User submits required documents.
3. System validates identity.
4. Verification status is updated.

Exit state: Verified user identity.

User Map

Platform Admin defines verification requirements. Organization Admin triggers verification for users. Instructors and learners submit verification data. Support users assist verification failures.

FEATURE 7: Consent & Privacy Management

Description

This feature manages user consent for data collection, processing, and usage.

It ensures transparency and legal compliance.

User Journey

1. User is presented with consent options.

2. User reviews and accepts or declines.
3. Consent is recorded.
4. System enforces consent choices.

Exit state: Consent-compliant usage.

User Map

Platform Admin defines consent policies. Organization Admin aligns organization-specific requirements. Instructors and learners manage their consent preferences. Support users view consent status for compliance.

FEATURE 8: Data Encryption & Key Management

Description

This feature ensures data is encrypted at rest and in transit using managed encryption keys.

It protects sensitive information from unauthorized access.

User Journey

1. Data is stored or transmitted.
2. Encryption is applied automatically.
3. Keys are securely managed and rotated.

Exit state: Data securely protected.

User Map

Platform Admin manages encryption policies and keys. Organization Admin does not access encryption settings. Instructors and learners are unaware of encryption mechanics. Support users operate under restricted visibility.

FEATURE 9: Audit Logs & Security Trails

Description

This feature records all critical actions for security, compliance, and investigation.

Logs are immutable and time-stamped.

User Journey

1. User performs a sensitive action.
2. System records the event.
3. Logs are stored securely.
4. Authorized users review logs.

Exit state: Action traceability ensured.

User Map

Platform Admin accesses full audit logs. Organization Admin reviews logs for their tenant. Instructors and learners do not access audit logs. Support users review logs during incidents.

FEATURE 10: Data Retention & Right-to-Erasure

Description

This feature enforces data retention rules and supports user requests to delete personal data.

It is critical for privacy compliance.

User Journey

1. Data reaches retention limit or deletion request is submitted.
2. System validates request.

3. Data is anonymized or erased.
4. Confirmation is recorded.

Exit state: Data compliance achieved.

User Map

Platform Admin defines retention rules. Organization Admin initiates erasure for users. Instructors and learners request data deletion. Support users assist processing.

FEATURE 11: Regulatory Compliance Framework

Description

This feature ensures Eduyata aligns with applicable regulations such as data protection and security standards.

It provides policy enforcement and evidence readiness.

User Journey

1. Compliance rules are configured.
2. System enforces controls.
3. Compliance reports are generated.
4. Audits are supported.

Exit state: Regulatory readiness.

User Map

Platform Admin manages compliance frameworks. Organization Admin ensures organizational adherence. Instructors and learners are subject to compliance controls. Support users assist audits.

FEATURE 12: Incident Detection & Response

Description

This feature detects security incidents and orchestrates response actions.

It minimizes damage and recovery time.

User Journey

1. Security event is detected.
2. Incident is classified.
3. Response actions are triggered.
4. Incident is resolved and documented.

Exit state: Incident contained.

User Map

Platform Admin leads incident response. Organization Admin is notified for tenant-related incidents. Instructors and learners may be impacted. Support users assist remediation.

FEATURE 13: Breach Notification & Reporting

Description

This feature manages mandatory notifications when data breaches occur.

It ensures transparency and legal compliance.

User Journey

1. Breach is confirmed.
2. Impact is assessed.

3. Notifications are generated.
4. Reports are archived.

Exit state: Breach reporting completed.

User Map

Platform Admin manages notifications and reporting. Organization Admin receives tenant-specific reports. Instructors and learners are notified if impacted. Support users assist communication.

FEATURE 14: Security Configuration & Policies

Description

This feature centralizes security-related configurations and enforcement policies.

It defines how strict the platform security posture is.

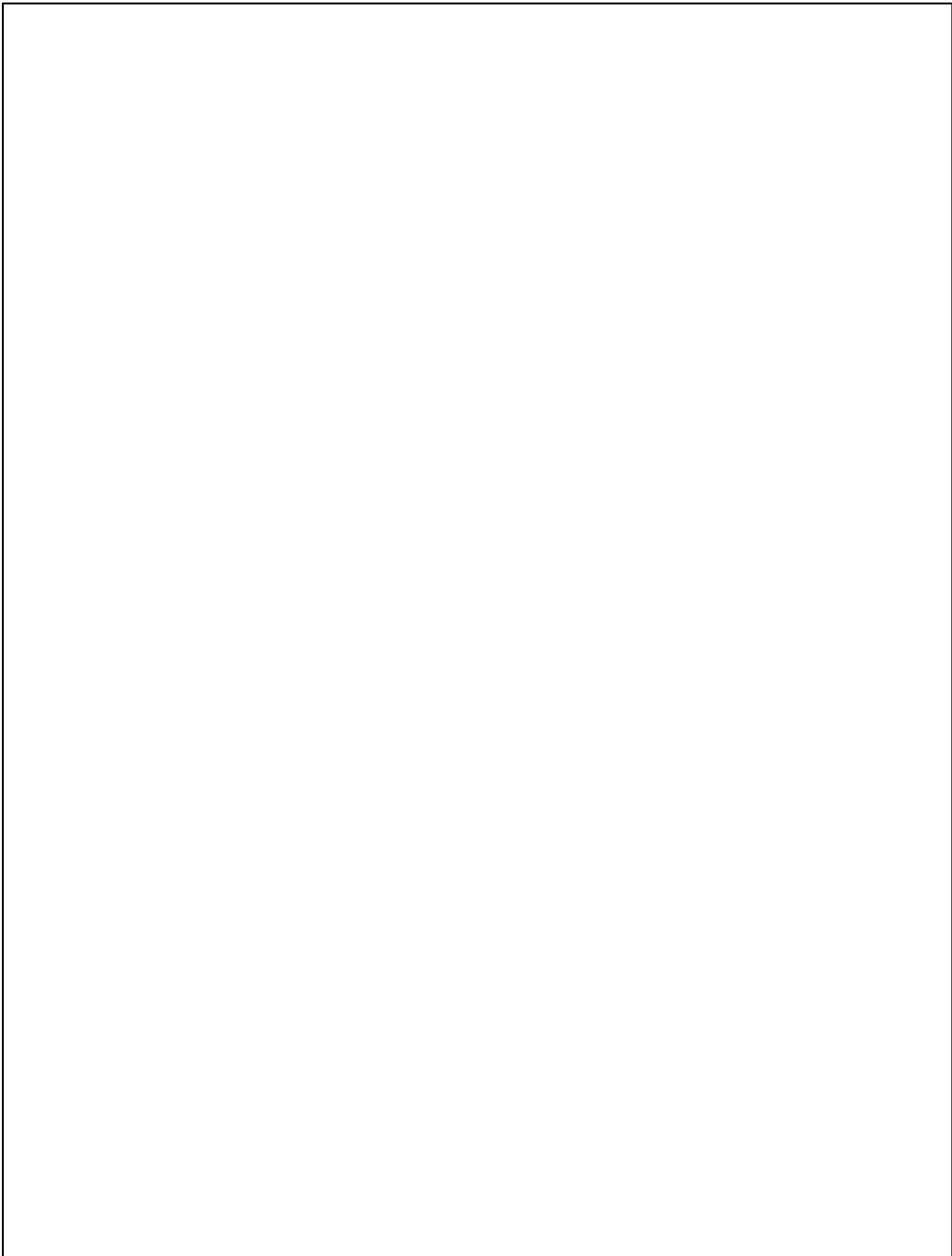
User Journey

1. Security policies are defined.
2. Policies are enforced system-wide.
3. Changes are audited.
4. Effects apply immediately.

Exit state: Security posture enforced.

User Map

Platform Admin defines and updates security policies. Organization Admin applies policies within allowed scope. Instructors and learners are governed by these policies. Support users follow defined security constraints.



EPIC 3 — Transaction, Billing & Entitlements Platform

FEATURE 1: Product Catalog & Pricing Plans

Description

This feature defines what Eduyata sells and how it is priced.

It manages products, plans, tiers, durations, and eligibility rules.

It is the single source of truth for monetization.

Nothing can be sold without this.

It does not process payments.

User Journey

1. Platform Admin creates products and pricing plans.
2. Plans are configured with duration, limits, and eligibility.
3. Products are published to learners or organizations.
4. Users view available plans during purchase flow.

Exit state: Purchasable offerings are available.

User Map

Platform Admin defines and manages all products and pricing plans. Organization Admin views plans applicable to their organization. Instructors and learners view available offerings. Support users reference catalog data for billing support.

FEATURE 2: Checkout & Payment Processing

Description

This feature handles the secure collection of payments.

It integrates with payment gateways to process transactions.

It must be reliable, auditable, and compliant.

It does not decide entitlement logic.

User Journey

1. User selects a product or plan.
2. User proceeds to checkout.
3. Payment details are entered.
4. Payment gateway processes transaction.
5. Success or failure is returned.

Exit state: Payment completed or failed.

User Map

Platform Admin configures payment gateways. Organization Admin may initiate payments for organizational purchases. Learners complete payments during checkout. Instructors usually do not initiate payments. Support users assist with payment issues.

FEATURE 3: Subscription Management

Description

This feature manages recurring purchases such as monthly or annual plans.

It controls renewals, upgrades, downgrades, and cancellations.

It ensures continuous access based on payment status.

User Journey

1. User purchases a subscription plan.
2. Subscription becomes active.
3. Renewals occur automatically.

4. User upgrades, downgrades, or cancels.
5. Subscription state updates accordingly.

Exit state: Subscription lifecycle managed.

User Map

Platform Admin defines subscription rules. Organization Admin manages subscriptions for their organization. Learners manage their own subscriptions. Instructors typically do not manage subscriptions. Support users assist subscription changes.

FEATURE 4: One-Time Purchases & Orders

Description

This feature supports non-recurring purchases such as individual courses or certifications.

It operates independently of subscriptions.

User Journey

1. User selects a one-time product.
2. Checkout is completed.
3. Order is created.
4. Payment is confirmed.
5. Entitlement is granted.

Exit state: Order completed.

User Map

Platform Admin configures one-time products. Organization Admin may purchase on behalf of users. Learners make one-time purchases. Instructors do not initiate purchases. Support users resolve order issues.

FEATURE 5: Wallet / Credits System

Description

This feature provides a stored-value system for prepaid credits.
Credits can be used instead of direct payment.

It improves flexibility and promotions.

User Journey

1. Credits are added to wallet.
2. User selects a product.
3. Wallet balance is applied.
4. Remaining balance is updated.

Exit state: Credits consumed correctly.

User Map

Platform Admin configures wallet rules. Organization Admin manages organizational credits.
Learners use wallet credits for purchases. Instructors do not manage wallets. Support users assist credit disputes.

FEATURE 6: Coupons, Discounts & Scholarships

Description

This feature applies price reductions based on eligibility.
It supports marketing and accessibility initiatives.

It must not break revenue accounting.

User Journey

1. Admin creates coupon or scholarship.
2. User enters code or qualifies automatically.
3. Discount is applied at checkout.
4. Final price is recalculated.

Exit state: Discounted purchase completed.

User Map

Platform Admin creates and controls discounts. Organization Admin applies scholarships within scope. Learners redeem coupons or receive scholarships. Instructors do not manage discounts. Support users assist redemption issues.

FEATURE 7: Entitlement Engine

Description

This feature determines what access a user gets after payment.
It connects billing outcomes to learning access.

It is the most critical link between money and value.

User Journey

1. Payment is confirmed.
2. Entitlement rules are evaluated.
3. Access is granted or revoked.
4. Entitlement duration is tracked.

Exit state: Correct access enforced.

User Map

Platform Admin defines entitlement rules. Organization Admin assigns entitlements to users. Learners receive access automatically. Instructors receive access only if entitled. Support users verify entitlement issues.

FEATURE 8: Usage Tracking & Metering

Description

This feature tracks consumption for usage-based billing models. It records how much of a service is used.

User Journey

1. User consumes a metered feature.
2. Usage is recorded.
3. Usage accumulates over time.
4. Billing references usage data.

Exit state: Accurate usage data stored.

User Map

Platform Admin defines metering rules. Organization Admin views usage reports. Learners generate usage passively. Instructors generate usage through activity. Support users review usage discrepancies.

FEATURE 9: Invoicing & Tax (GST) Management

Description

This feature generates invoices and calculates taxes. It ensures compliance with local tax regulations such as GST.

User Journey

1. Transaction is completed.
2. Tax is calculated.
3. Invoice is generated.
4. Invoice is delivered.

Exit state: Compliant invoice issued.

User Map

Platform Admin configures tax rules. Organization Admin receives invoices. Learners receive invoices for purchases. Instructors do not handle invoices. Support users assist invoice corrections.

FEATURE 10: Payment Failure, Retry & Dunning

Description

This feature handles failed payments and recovery attempts.
It reduces involuntary churn.

User Journey

1. Payment attempt fails.
2. Retry logic is triggered.
3. User is notified.
4. Subscription is paused or resumed.

Exit state: Payment recovered or access suspended.

User Map

Platform Admin defines retry policies. Organization Admin monitors failures. Learners update payment methods. Instructors are not involved. Support users assist recovery.

FEATURE 11: Refunds & Chargebacks

Description

This feature manages payment reversals and disputes.
It protects both users and the platform.

User Journey

1. Refund or dispute is initiated.
2. Transaction is reviewed.
3. Refund or chargeback is processed.
4. Entitlement is adjusted.

Exit state: Financial reversal completed.

User Map

Platform Admin approves refunds. Organization Admin requests refunds. Learners initiate refund requests. Instructors are not involved. Support users manage disputes.

FEATURE 12: Revenue & Transaction Ledger

Description

This feature records all financial movements in an immutable ledger.
It is required for reconciliation and audits.

User Journey

1. Transaction occurs.
2. Ledger entry is created.
3. Entry is locked.
4. Reports reference ledger data.

Exit state: Financial traceability ensured.

User Map

Platform Admin accesses full ledger. Organization Admin views organization-specific records. Learners do not access ledger data. Instructors do not access ledger data. Support users view limited records.

FEATURE 13: Billing Notifications & Receipts

Description

This feature communicates billing events to users.
It ensures transparency and trust.

User Journey

1. Billing event occurs.
2. Notification is triggered.
3. Receipt is generated.
4. User receives confirmation.

Exit state: User informed of billing activity.

User Map

Platform Admin configures notification rules. Organization Admin receives billing alerts. Learners receive receipts and alerts. Instructors are not notified. Support users reference notifications.

FEATURE 14: Admin Billing Console

Description

This feature provides administrators with centralized billing control. It exposes billing data, actions, and controls.

User Journey

1. Admin opens billing console.
2. Views transactions and subscriptions.
3. Performs billing actions.
4. Changes take effect immediately.

Exit state: Billing managed centrally.

User Map

Platform Admin fully operates the billing console. Organization Admin manages billing within their organization. Learners and instructors do not access the billing console. Support users access limited views for troubleshooting.

EPIC 4 — Platform Services & Integrations

FEATURE 1: Public API Framework

Description

This feature exposes Eduyata functionality through well-defined APIs.

It allows external systems to interact with Eduyata in a controlled, versioned manner.

It is the foundation for all integrations.

Without this, the platform is closed.

User Journey

1. Platform Admin enables API access.
2. Developer or system initiates API requests.
3. Requests are validated and routed.
4. Responses are returned with enforced policies.

Exit state: External systems can interact with Eduyata safely.

User Map

Platform Admin governs API availability and policies. Organization Admin may enable APIs for their tenant. Developers and external systems consume APIs. Instructors and learners do not interact directly. Support users assist API issue resolution.

FEATURE 2: API Authentication & Access Keys

Description

This feature secures API access using keys or tokens.

It ensures only authorized systems can call Eduyata APIs.

It is mandatory for API usage.

User Journey

1. Admin generates API credentials.
2. Credentials are securely stored.
3. API request includes credentials.
4. System validates credentials.
5. Request is accepted or rejected.

Exit state: Authenticated API access enforced.

User Map

Platform Admin creates and rotates API keys. Organization Admin manages keys for their integrations. External developers use keys for API access. Instructors and learners are not involved. Support users assist with key issues.

FEATURE 3: Webhook Framework

Description

This feature allows Eduyata to send real-time events to external systems.

It enables reactive integrations.

It does not receive data.

It only pushes events.

User Journey

1. Admin configures webhook endpoint.

2. Event occurs in Eduyata.
3. Webhook payload is generated.
4. Payload is delivered to external system.
5. Delivery status is logged.

Exit state: External systems receive events.

User Map

Platform Admin configures webhook rules. Organization Admin enables webhooks for their tenant. External systems receive webhook events. Instructors and learners do not interact. Support users troubleshoot delivery failures.

FEATURE 4: Integration Marketplace / Catalog

Description

This feature lists available integrations that can be connected to Eduyata.

It simplifies discovery and activation of integrations.

User Journey

1. Admin browses integration catalog.
2. Selects an integration.
3. Reviews capabilities and permissions.
4. Activates integration.

Exit state: Integration enabled.

User Map

Platform Admin curates the marketplace. Organization Admin activates integrations. External partners publish integrations. Instructors and learners indirectly benefit. Support users assist activation.

FEATURE 5: Third-Party App Connectors

Description

This feature provides prebuilt connectors to common third-party applications.

It reduces integration effort.

User Journey

1. Admin selects connector.
2. Credentials are configured.
3. Connection is tested.
4. Data exchange begins.

Exit state: Connected third-party application.

User Map

Platform Admin maintains connectors. Organization Admin configures connectors. External apps exchange data. Instructors and learners are not directly involved. Support users troubleshoot connector issues.

FEATURE 6: SSO & Identity Integrations

Description

This feature integrates Eduyata with external identity providers and directories.

It supports enterprise authentication and identity sync.

User Journey

1. Admin configures identity provider.
2. Trust relationship is established.
3. Users authenticate via provider.
4. Identity is synchronized.

Exit state: External identity integrated.

User Map

Platform Admin manages identity integrations. Organization Admin enables identity sync. Instructors and learners authenticate via integrated providers. Support users assist identity issues.

FEATURE 7: Data Import & Export Tools

Description

This feature enables bulk data movement into and out of Eduyata.

It supports migrations, reporting, and backups.

User Journey

1. Admin selects import or export.
2. Data format is chosen.
3. Process is executed.
4. Results are validated.

Exit state: Data transferred successfully.

User Map

Platform Admin manages data tools. Organization Admin imports or exports organization data. Instructors and learners do not access bulk data tools. Support users assist data issues.

FEATURE 8: Event Bus / Messaging Layer

Description

This feature enables internal event-driven communication between services.

It decouples system components.

User Journey

1. System event occurs.
2. Event is published to bus.
3. Subscribed services receive event.
4. Actions are triggered asynchronously.

Exit state: Event-driven processing completed.

User Map

Platform Admin oversees messaging infrastructure. Developers and services consume events. End users are not directly involved. Support users monitor failures.

FEATURE 9: SDKs & Developer Tooling

Description

This feature provides libraries and tools for developers to integrate faster.

It improves developer experience and adoption.

User Journey

1. Developer downloads SDK.
2. SDK is integrated into application.
3. APIs are called via SDK.
4. Responses are handled.

Exit state: Faster integration development.

User Map

Platform Admin publishes SDKs. External developers use SDKs. Organization Admin may sponsor development. Instructors and learners are not involved. Support users assist developer issues.

FEATURE 10: Integration Configuration & Secrets Vault

Description

This feature securely stores credentials and secrets used by integrations.

It prevents credential leakage.

User Journey

1. Admin stores secrets.
2. Secrets are encrypted.
3. Integrations retrieve secrets securely.
4. Access is audited.

Exit state: Secrets protected.

User Map

Platform Admin manages secrets vault. Organization Admin stores tenant secrets. External systems never see raw secrets. Instructors and learners are not involved. Support users have restricted access.

FEATURE 11: Rate Limits & Quotas for APIs

Description

This feature enforces limits on API usage.

It protects platform stability.

User Journey

1. API request is received.
2. Rate limits are checked.
3. Request is allowed or throttled.
4. Usage metrics are updated.

Exit state: API usage controlled.

User Map

Platform Admin defines rate limits. Organization Admin monitors usage. External developers adjust usage patterns. Instructors and learners are not involved. Support users assist quota issues.

FEATURE 12: Unified Notification & Messaging Service

Description

This feature centralizes all system notifications across channels.

It ensures consistent communication.

User Journey

1. Event triggers notification.
2. Message is generated.
3. Delivery channel is selected.
4. Notification is sent.

Exit state: User or system notified.

User Map

Platform Admin configures notification rules. Organization Admin customizes notifications. Instructors and learners receive notifications. Support users review delivery issues.

FEATURE 13: Integration Monitoring & Logs

Description

This feature provides visibility into integration health and failures.

It is critical for operational reliability.

User Journey

1. Integration runs.
2. Metrics and logs are captured.
3. Failures are detected.
4. Alerts are generated.

Exit state: Integration health monitored.

User Map

Platform Admin monitors all integrations. Organization Admin monitors tenant integrations. External developers debug their integrations. Instructors and learners are not involved. Support users investigate incidents.

EPIC 5 — Platform Operations, Admin & Analytics

FEATURE 1: Unified Admin Console

Description

This feature provides a single control center for operating the Eduyata platform.

It consolidates administration, configuration, monitoring, and operational actions.

It is the primary interface for platform and organization administrators.

User Journey

1. Admin logs into Eduyata.
2. Admin opens the unified admin console.
3. Relevant modules load based on admin role.
4. Admin performs configuration or operational tasks.
5. Changes are applied and audited.

Exit state: Administrative actions executed centrally.

User Map

Platform Admin fully operates the unified admin console. Organization Admin accesses organization-scoped controls. Instructors and learners do not access the admin console. Support users access limited operational views.

FEATURE 2: Product & Tenant Configuration Management

Description

This feature manages product-level and tenant-level configuration settings.

It controls how Eduyata behaves for different organizations and products.

User Journey

1. Admin selects product or tenant.
2. Configuration options are loaded.
3. Admin updates settings.
4. System validates changes.
5. Configuration is applied.

Exit state: Updated configuration in effect.

User Map

Platform Admin manages global product configurations. Organization Admin manages tenant-specific settings. Instructors and learners are affected by configurations but cannot modify them. Support users view configurations for troubleshooting.

FEATURE 3: Feature Flags & Experiments

Description

This feature enables controlled rollout of new features and experiments.

It supports gradual releases, testing, and rollback.

User Journey

1. Admin creates a feature flag or experiment.
2. Target audience is defined.
3. Feature is enabled for selected users.
4. Metrics are observed.
5. Feature is rolled out or disabled.

Exit state: Feature safely released or reverted.

User Map

Platform Admin manages feature flags globally. Organization Admin may enable features within allowed scope. Instructors and learners experience features based on targeting. Support users assist in rollout validation.

FEATURE 4: Approval & Moderation Workflows

Description

This feature enforces review and approval for sensitive actions and content.

It prevents unreviewed changes from going live.

User Journey

1. User submits item for approval.
2. System routes request to approver.
3. Approver reviews submission.
4. Request is approved or rejected.
5. Outcome is recorded.

Exit state: Controlled approval decision applied.

User Map

Platform Admin defines approval rules. Organization Admin acts as approver for organization-level actions. Instructors submit items for approval. Learners do not participate. Support users monitor workflow issues.

FEATURE 5: Support, Ticketing & Escalation Tools

Description

This feature enables structured handling of user issues and requests.

It ensures timely resolution through defined escalation paths.

User Journey

1. User raises a support ticket.
2. Ticket is categorized and assigned.
3. Support investigates the issue.
4. Issue is resolved or escalated.
5. Ticket is closed.

Exit state: User issue resolved.

User Map

Platform Admin oversees support operations. Organization Admin raises and tracks tickets. Instructors and learners submit support requests. Support users manage tickets and escalations.

FEATURE 6: Dispute & Grievance Management

Description

This feature manages formal disputes and grievances raised by users.

It ensures fairness, traceability, and resolution accountability.

User Journey

1. User submits grievance.

2. Case is registered.
3. Investigation is conducted.
4. Resolution is decided.
5. User is notified.

Exit state: Dispute resolved and documented.

User Map

Platform Admin defines grievance processes. Organization Admin handles organization-level disputes. Instructors and learners raise grievances. Support users assist investigation.

FEATURE 7: System Health, Monitoring & Logs

Description

This feature provides visibility into platform performance and stability.

It is essential for reliability and uptime.

User Journey

1. System metrics are collected.
2. Health dashboards are updated.
3. Alerts trigger on anomalies.
4. Admin takes corrective action.

Exit state: System health maintained.

User Map

Platform Admin monitors system health continuously. Organization Admin may view limited health indicators. Instructors and learners are not involved. Support users respond to incidents.

FEATURE 8: Usage Analytics & Funnels

Description

This feature tracks how users interact with the platform.

It identifies drop-offs, engagement patterns, and adoption trends.

User Journey

1. User actions are captured.
2. Data flows into analytics pipelines.
3. Funnels and metrics are generated.
4. Insights are reviewed.

Exit state: Actionable usage insights available.

User Map

Platform Admin accesses full analytics. Organization Admin views organization-specific analytics. Instructors view limited engagement insights. Learners do not access analytics. Support users reference analytics during investigations.

FEATURE 9: Business & Revenue Dashboards

Description

This feature presents financial and business performance metrics.

It supports strategic and operational decision-making.

User Journey

1. Transactions and usage data are aggregated.

2. Dashboards are generated.
3. Metrics are visualized.
4. Admin reviews performance.

Exit state: Business performance visibility.

User Map

Platform Admin views all business dashboards. Organization Admin views revenue related to their organization. Instructors and learners do not access business dashboards. Support users reference data for billing issues.

FEATURE 10: Data Management, Backup & Restore

Description

This feature ensures data durability and recoverability.

It protects against data loss and corruption.

User Journey

1. Data is backed up on schedule.
2. Backup integrity is verified.
3. Restore is initiated if required.
4. Data is recovered.

Exit state: Data safely restored.

User Map

Platform Admin manages backups and restores. Organization Admin does not initiate restores. Instructors and learners are unaware of backup operations. Support users assist recovery under admin direction.

FEATURE 11: Role-Based Admin Access Control

Description

This feature restricts administrative actions based on admin roles.

It prevents over-privileged access.

User Journey

1. Admin role is assigned.
2. Permissions are evaluated.
3. Admin accesses allowed actions.
4. Unauthorized actions are blocked.

Exit state: Secure admin access enforced.

User Map

Platform Admin defines admin roles. Organization Admin operates within assigned permissions. Instructors and learners are not admins. Support users operate under delegated admin roles.

FEATURE 12: Content & Data Quality Governance

Description

This feature enforces quality standards for content and data.

It ensures accuracy, consistency, and reliability.

User Journey

1. Content or data is submitted.
2. Quality rules are applied.
3. Issues are flagged.
4. Corrections are made or approved.

Exit state: Quality-compliant content and data.

User Map

Platform Admin defines governance rules. Organization Admin enforces quality within their scope. Instructors submit content for review. Learners consume governed content. Support users assist enforcement.

FEATURE 13: Platform Configuration APIs

Description

This feature exposes administrative configuration via APIs.

It enables automation and integration of platform operations.

User Journey

1. API credentials are issued.
2. Configuration API is called.
3. Request is validated.
4. Configuration is updated.

Exit state: Configuration automated successfully.

User Map

Platform Admin uses configuration APIs. Organization Admin may use APIs within scope. Instructors and learners do not access configuration APIs. Support users assist API troubleshooting.

FEATURE 14: Localization & Globalization Management

Description

This feature enables Eduyata to support multiple languages, regions, and formats.

It ensures global usability and compliance.

User Journey

1. Localization settings are defined.
2. Language and regional formats are applied.
3. User selects preferred language.
4. Platform adapts accordingly.

Exit state: Localized user experience delivered.

User Map

Platform Admin manages global localization settings. Organization Admin selects supported languages. Instructors and learners use localized interfaces. Support users assist localization issues.

EPIC 6 — Learning & Assessment Core Engine

FEATURE 1: Course & Program Management

Description

This feature allows creation and management of courses and higher-level programs.

Courses represent individual learning units. Programs group multiple courses into structured offerings.

It is the backbone of all learning delivery.

User Journey

1. Admin or instructor creates a course or program.
2. Metadata, structure, and visibility are defined.
3. Course or program is published.
4. Learners can discover and enroll.

Exit state: Course or program available for learning.

User Map

Platform Admin governs global course standards. Organization Admin manages courses within their organization. Instructors create and manage their assigned courses. Learners consume published courses. Support users assist with course issues.

FEATURE 2: Modular Learning Structure

Description

This feature breaks courses into modules, units, and lessons.

It enables flexible and scalable content organization.

User Journey

1. Instructor defines course structure.
2. Modules and lessons are added.
3. Order and visibility are set.
4. Learners progress module by module.

Exit state: Structured learning flow established.

User Map

Platform Admin defines structural rules. Organization Admin enforces standards. Instructors design modular structures. Learners navigate modules. Support users assist structure-related issues.

FEATURE 3: Curriculum & Syllabus Framework

Description

This feature defines formal curricula and syllabi aligned to educational standards.

It ensures academic consistency and compliance.

User Journey

1. Curriculum framework is defined.
2. Syllabus is mapped to courses.
3. Curriculum is approved.
4. Courses follow defined syllabus.

Exit state: Standardized curriculum applied.

User Map

Platform Admin defines curriculum frameworks. Organization Admin approves syllabi. Instructors align content to syllabus. Learners follow curriculum structure. Support users assist compliance issues.

FEATURE 4: Course Lifecycle Management

Description

This feature manages course states from draft to retirement.

It prevents outdated or invalid courses from being used.

User Journey

1. Course is created in draft state.
2. Course is reviewed and published.
3. Course runs actively.
4. Course is archived or retired.

Exit state: Course lifecycle correctly enforced.

User Map

Platform Admin defines lifecycle rules. Organization Admin controls course status. Instructors update course content. Learners access only active courses. Support users assist lifecycle transitions.

FEATURE 5: Skill & Talent Taxonomy

Description

This feature defines a structured skill hierarchy.

It creates a common language for skills across the platform.

User Journey

1. Skill taxonomy is defined.
2. Skills are categorized and leveled.
3. Taxonomy is published.
4. Other features reference skills.

Exit state: Standard skill framework available.

User Map

Platform Admin defines global skill taxonomy. Organization Admin may extend taxonomy. Instructors reference skills in courses. Learners view skill alignment. Support users assist taxonomy maintenance.

FEATURE 6: Skill-to-Course Mapping

Description

This feature maps courses to skills they teach or reinforce.

It enables outcome-based learning.

User Journey

1. Instructor selects relevant skills.
2. Skills are mapped to course or module.
3. Mapping is validated.
4. Learner progress updates skill indicators.

Exit state: Skills linked to learning content.

User Map

Platform Admin governs mapping rules. Organization Admin reviews mappings. Instructors map skills to courses. Learners gain skill visibility. Support users assist mapping issues.

FEATURE 7: Competency & Mastery Model

Description

This feature defines how skill mastery is measured.

It supports levels such as beginner, intermediate, and advanced.

User Journey

1. Competency levels are defined.
2. Assessment criteria are linked.
3. Learner performance is evaluated.
4. Mastery level is updated.

Exit state: Learner competency tracked.

User Map

Platform Admin defines mastery models. Organization Admin adopts models. Instructors assess competency. Learners track mastery progression. Support users assist evaluation issues.

FEATURE 8: Enrollment Rules & Eligibility

Description

This feature controls who can enroll in courses or programs.

It enforces eligibility, limits, and constraints.

User Journey

1. Enrollment rules are defined.
2. Learner attempts enrollment.
3. Eligibility is checked.
4. Enrollment succeeds or fails.

Exit state: Enrollment rules enforced.

User Map

Platform Admin defines enrollment policies. Organization Admin sets organization-specific rules. Instructors may recommend enrollment. Learners enroll based on eligibility. Support users assist enrollment issues.

FEATURE 9: Progress & Completion Tracking

Description

This feature tracks learner progress and completion status.

It ensures accurate reporting of learning outcomes.

User Journey

1. Learner starts course.
2. Progress events are recorded.
3. Completion criteria are met.
4. Course is marked complete.

Exit state: Learning progress recorded.

User Map

Platform Admin defines tracking rules. Organization Admin views progress reports. Instructors monitor learner progress. Learners track their own progress. Support users assist progress discrepancies.

FEATURE 10: Learning Paths & Programs

Description

This feature sequences courses into guided learning paths.

It supports structured skill development journeys.

User Journey

1. Learning path is created.
2. Courses are ordered.
3. Learner enrolls in path.
4. Progress flows sequentially.

Exit state: Guided learning journey established.

User Map

Platform Admin defines path frameworks. Organization Admin manages paths. Instructors contribute courses. Learners follow paths. Support users assist navigation issues.

FEATURE 11: Prerequisite & Dependency Engine

Description

This feature enforces dependencies between courses or modules.

It prevents learners from skipping required learning.

User Journey

1. Prerequisites are defined.
2. Learner attempts access.
3. System checks dependencies.
4. Access is allowed or blocked.

Exit state: Dependencies enforced.

User Map

Platform Admin defines dependency rules. Organization Admin configures prerequisites. Instructors align content sequencing. Learners unlock content progressively. Support users assist access issues.

FEATURE 12: Outcome & Credit System

Description

This feature assigns outcomes or academic credits to learning activities.

It supports formal recognition of learning.

User Journey

1. Credit rules are defined.
2. Learner completes activity.
3. Credits are awarded.
4. Credits are recorded.

Exit state: Outcomes and credits assigned.

User Map

Platform Admin defines credit systems. Organization Admin maps credits to programs. Instructors deliver credit-bearing courses. Learners earn credits. Support users assist credit disputes.

FEATURE 13: Certificate Eligibility Logic

Description

This feature determines when a learner qualifies for certificates.

It enforces strict eligibility criteria.

User Journey

1. Learner completes requirements.
2. Eligibility is evaluated.
3. Certificate is unlocked.
4. Learner receives certificate.

Exit state: Certificate eligibility enforced.

User Map

Platform Admin defines certificate rules. Organization Admin approves certificate issuance. Instructors validate completion. Learners receive certificates. Support users assist certificate issues.

FEATURE 14: Gamification & Rewards Engine

Description

This feature motivates learners through points, badges, and rewards.

It enhances engagement without affecting academic integrity.

User Journey

1. Gamification rules are defined.
2. Learner performs actions.
3. Rewards are triggered.
4. Progress and rewards are displayed.

Exit state: Engagement reinforced.

User Map

Platform Admin defines gamification framework. Organization Admin enables or disables gamification. Instructors align activities with rewards. Learners earn and view rewards. Support users assist reward discrepancies.

EPIC 7 — Learning Experience, Gamification & Engagement

FEATURE 1: Enrollment & Onboarding Workflows

Description

This feature manages how learners enter courses, programs, or learning paths and how they are guided at the start.

It ensures learners understand expectations, structure, and next steps.

It focuses on clarity and reduction of drop-offs.

User Journey

1. Learner selects a course or program.
2. Enrollment eligibility is checked.
3. Enrollment is confirmed.
4. Onboarding content is shown.
5. Learner is guided to the first lesson.

Exit state: Learner successfully onboarded.

User Map

Platform Admin defines onboarding standards. Organization Admin customizes onboarding flows. Instructors configure course-specific onboarding. Learners go through enrollment and onboarding. Support users assist onboarding issues.

FEATURE 2: Lesson Delivery & Navigation

Description

This feature delivers learning content and controls how learners move through lessons.

It ensures a consistent and intuitive learning flow.

User Journey

1. Learner opens a lesson.
2. Content loads based on permissions.
3. Learner navigates through lesson sections.
4. Completion is recorded.

Exit state: Lesson consumed and tracked.

User Map

Platform Admin defines delivery rules. Organization Admin enforces navigation standards. Instructors design lesson flow. Learners consume lessons. Support users assist access or navigation issues.

FEATURE 3: Practice & Activity Engine

Description

This feature enables hands-on practice through exercises, simulations, or activities.

It reinforces learning through action.

User Journey

1. Learner starts a practice activity.
2. Activity instructions are shown.
3. Learner completes the activity.
4. System evaluates responses.

Exit state: Practice attempt completed.

User Map

Platform Admin defines activity frameworks. Organization Admin enables activity types. Instructors create practice activities. Learners perform activities. Support users assist activity issues.

FEATURE 4: Practice Session Logging

Description

This feature records detailed data about learner practice sessions.

It supports analytics, feedback, and improvement tracking.

User Journey

1. Learner begins practice.
2. Session data is captured.
3. Practice ends.
4. Logs are stored and analyzed.

Exit state: Practice data recorded.

User Map

Platform Admin defines logging policies. Organization Admin views aggregated logs. Instructors review learner sessions. Learners view their own practice history. Support users assist data discrepancies.

FEATURE 5: Assignments & Submissions

Description

This feature manages assignment creation, submission, and tracking.

It supports structured evaluation of learner work.

User Journey

1. Instructor creates an assignment.
2. Learner views assignment details.
3. Learner submits work.
4. Submission is recorded.

Exit state: Assignment submitted.

User Map

Platform Admin defines assignment rules. Organization Admin enforces policies. Instructors create and manage assignments. Learners submit work. Support users assist submission issues.

FEATURE 6: Assessments & Quizzes

Description

This feature evaluates learner understanding through quizzes and assessments.

It supports both formative and summative evaluation.

User Journey

1. Learner starts assessment.
2. Questions are presented.
3. Learner submits answers.
4. Assessment is graded.

Exit state: Assessment completed and scored.

User Map

Platform Admin defines assessment frameworks. Organization Admin configures assessment policies. Instructors create assessments. Learners take assessments. Support users assist assessment issues.

FEATURE 7: Feedback & Grading Workflows

Description

This feature enables instructors to grade work and provide feedback.

It ensures timely and structured learner evaluation.

User Journey

1. Instructor reviews submission.
2. Grade and feedback are provided.
3. Learner receives notification.
4. Feedback is acknowledged.

Exit state: Feedback delivered.

User Map

Platform Admin defines grading standards. Organization Admin enforces grading policies. Instructors grade and provide feedback. Learners receive feedback. Support users assist grading disputes.

FEATURE 8: Evidence Upload (Media)

Description

This feature allows learners to upload supporting evidence such as documents, images, audio, or video.

It supports practical and project-based learning.

User Journey

1. Learner selects upload option.
2. Media is uploaded.
3. Validation checks run.
4. Evidence is attached to activity.

Exit state: Evidence successfully uploaded.

User Map

Platform Admin defines upload policies. Organization Admin manages storage limits. Instructors review submitted evidence. Learners upload media. Support users assist upload issues.

FEATURE 9: Progress Reviews & Milestones

Description

This feature tracks milestone achievements and supports progress reviews.

It helps learners and instructors assess advancement.

User Journey

1. Learner reaches milestone.

2. System records achievement.
3. Review is conducted if required.
4. Progress is updated.

Exit state: Milestone achieved.

User Map

Platform Admin defines milestone frameworks. Organization Admin monitors progress. Instructors conduct reviews. Learners track milestones. Support users assist review issues.

FEATURE 10: Certificates Issuance Workflow

Description

This feature manages the issuance of certificates after eligibility is met.

It ensures certificates are issued accurately and securely.

User Journey

1. Learner completes requirements.
2. Eligibility is verified.
3. Certificate is generated.
4. Learner receives certificate.

Exit state: Certificate issued.

User Map

Platform Admin defines issuance rules. Organization Admin approves certificates. Instructors validate completion. Learners receive certificates. Support users assist certificate issues.

FEATURE 11: Study Groups & Learning Circles

Description

This feature enables collaborative learning through small groups.

It encourages peer interaction and accountability.

User Journey

1. Study group is created.
2. Learners join group.
3. Group activities are conducted.
4. Participation is tracked.

Exit state: Collaborative learning active.

User Map

Platform Admin defines group policies. Organization Admin enables study groups. Instructors facilitate groups. Learners participate in groups. Support users assist group management.

FEATURE 12: Peer Feedback & Reviews

Description

This feature allows learners to review and provide feedback on peer work.

It builds reflection and collaborative evaluation skills.

User Journey

1. Peer review is assigned.
2. Learner reviews peer submission.

3. Feedback is submitted.
4. Feedback is shared.

Exit state: Peer review completed.

User Map

Platform Admin defines peer review rules. Organization Admin enables peer feedback. Instructors moderate reviews. Learners give and receive feedback. Support users assist moderation issues.

FEATURE 13: Notifications & Reminders

Description

This feature delivers timely notifications and reminders related to learning activities.

It keeps learners engaged and on track.

User Journey

1. Trigger event occurs.
2. Notification is generated.
3. Delivery channel is selected.
4. Notification is delivered.

Exit state: User informed.

User Map

Platform Admin defines notification rules. Organization Admin customizes notifications. Instructors trigger course notifications. Learners receive reminders. Support users troubleshoot delivery issues.

FEATURE 14: Engagement Tracking & Signals

Description

This feature tracks engagement indicators such as activity frequency and responsiveness.

It helps identify disengaged or highly engaged learners.

User Journey

1. Learner activity occurs.
2. Engagement signals are captured.
3. Signals are analyzed.
4. Insights are generated.

Exit state: Engagement insights available.

User Map

Platform Admin defines engagement metrics. Organization Admin monitors engagement trends. Instructors track learner engagement. Learners generate engagement signals through activity. Support users assist analytics issues.

EPIC 8 — Learning Content & Knowledge Layer

FEATURE 1: Central Content Repository

Description

This feature provides a single, authoritative store for all learning content in Eduyata.

It eliminates content sprawl and ensures consistency across courses, programs, and products.

It is the backbone of the knowledge layer.

User Journey

1. Content is created or uploaded into the repository.
2. Content is stored with unique identifiers.
3. Other features reference content from the repository.
4. Updates propagate wherever the content is used.

Exit state: Centralized and reusable content available.

User Map

Platform Admin governs repository standards. Organization Admin manages organization content. Instructors store and reuse content. Learners consume content. Support users assist repository issues.

FEATURE 2: Multi-Format Content Support

Description

This feature allows the platform to support multiple content formats such as video, audio, documents, images, and links.

It ensures flexibility in content creation and delivery.

User Journey

1. Instructor selects content format.
2. Content is uploaded or linked.
3. Format validation is performed.
4. Content becomes available for use.

Exit state: Multi-format content supported.

User Map

Platform Admin defines supported formats. Organization Admin enforces format policies. Instructors upload content in supported formats. Learners view content. Support users assist format-related issues.

FEATURE 3: Content Upload & Authoring Tools

Description

This feature provides tools to upload, create, and edit learning content.

It enables instructors to author content directly within the platform.

User Journey

1. Instructor opens authoring tool.
2. Content is created or uploaded.
3. Edits are made and previewed.
4. Content is saved or published.

Exit state: Authored content ready for use.

User Map

Platform Admin defines authoring capabilities. Organization Admin enables authoring access. Instructors create and edit content. Learners do not author content. Support users assist authoring issues.

FEATURE 4: Content Organization & Foldering

Description

This feature enables structured organization of content using folders and hierarchies.

It improves content discoverability and governance.

User Journey

1. Folder structure is created.
2. Content is placed into folders.
3. Hierarchy is maintained.
4. Users browse organized content.

Exit state: Content logically organized.

User Map

Platform Admin defines organization standards. Organization Admin manages folder structures. Instructors organize their content. Learners browse organized content. Support users assist navigation issues.

FEATURE 5: Metadata, Tags & Taxonomy

Description

This feature enriches content with metadata, tags, and taxonomies.

It enables powerful search, filtering, and recommendation.

User Journey

1. Metadata fields are defined.
2. Content is tagged.
3. Taxonomy is applied.
4. Metadata is indexed.

Exit state: Content enriched for discovery.

User Map

Platform Admin defines metadata standards. Organization Admin manages taxonomies. Instructors tag content. Learners benefit from improved discovery. Support users assist metadata issues.

FEATURE 6: Content Versioning & Reuse

Description

This feature tracks content changes and allows reuse across multiple courses.

It prevents duplication and supports rollback.

User Journey

1. Content is edited.
2. New version is created.
3. Previous versions are retained.
4. Reused content updates propagate.

Exit state: Versioned, reusable content maintained.

User Map

Platform Admin defines versioning rules. Organization Admin manages reuse policies. Instructors update and reuse content. Learners always see the correct version. Support users assist rollback scenarios.

FEATURE 7: Access Control & Visibility Rules

Description

This feature controls who can view or use content.

It enforces permissions based on role, tenant, and context.

User Journey

1. Visibility rules are defined.
2. User requests content access.
3. Access rules are evaluated.
4. Content is shown or hidden.

Exit state: Secure content access enforced.

User Map

Platform Admin defines access policies. Organization Admin sets visibility rules. Instructors control content sharing. Learners see permitted content only. Support users assist access issues.

FEATURE 8: Search & Discovery Engine

Description

This feature enables users to search and discover content efficiently.

It uses metadata, tags, and relevance ranking.

User Journey

1. User enters search query.
2. Search engine processes query.
3. Results are ranked and displayed.
4. User selects content.

Exit state: Relevant content discovered.

User Map

Platform Admin configures search behavior. Organization Admin manages discovery scope. Instructors search and reuse content. Learners discover learning materials. Support users assist search issues.

FEATURE 9: Download & Offline Access

Description

This feature allows controlled downloading of content for offline use.

It balances accessibility with content protection.

User Journey

1. User selects download option.
2. Eligibility is checked.
3. Content is downloaded securely.
4. Offline access is tracked.

Exit state: Content available offline.

User Map

Platform Admin defines offline policies. Organization Admin controls download permissions. Instructors enable offline access if allowed. Learners download content when permitted. Support users assist offline issues.

FEATURE 10: Notes, Bookmarks & Highlights

Description

This feature allows learners to annotate content with notes, bookmarks, and highlights.

It supports personalized learning.

User Journey

1. Learner opens content.
2. Note or highlight is added.
3. Bookmark is saved.
4. Annotations are retrieved later.

Exit state: Personalized annotations saved.

User Map

Platform Admin defines annotation policies. Organization Admin enables annotation features. Instructors view annotations if shared. Learners create notes and highlights. Support users assist annotation issues.

FEATURE 11: Reference Libraries & Resources

Description

This feature provides curated collections of reference materials.

It supports supplementary and self-directed learning.

User Journey

1. Reference library is created.
2. Resources are added.
3. Library is published.
4. Learners browse resources.

Exit state: Reference materials accessible.

User Map

Platform Admin defines library standards. Organization Admin curates resources. Instructors contribute references. Learners access libraries. Support users assist resource issues.

FEATURE 12: Content Quality & Validation Rules

Description

This feature enforces quality checks before content is published.

It ensures accuracy, completeness, and compliance.

User Journey

1. Content is submitted for validation.
2. Quality rules are applied.
3. Issues are flagged.
4. Content is approved or rejected.

Exit state: Quality-controlled content published.

User Map

Platform Admin defines quality rules. Organization Admin enforces validation. Instructors submit content for review. Learners consume approved content. Support users assist validation issues.

FEATURE 13: Content Analytics & Usage Insights

Description

This feature tracks how content is consumed and used.

It informs improvement and optimization decisions.

User Journey

1. Content is accessed.
2. Usage events are recorded.
3. Analytics are generated.
4. Insights are reviewed.

Exit state: Content usage insights available.

User Map

Platform Admin accesses global analytics. Organization Admin views organization insights. Instructors analyze content performance. Learners do not access analytics. Support users assist data interpretation.

FEATURE 14: Archival & Lifecycle Management

Description

This feature manages content lifecycle from creation to archival.

It prevents outdated content from remaining active.

User Journey

1. Content reaches end-of-life.
2. Archival rules are applied.
3. Content is archived or retired.
4. Access is restricted or removed.

Exit state: Content lifecycle enforced.

User Map

Platform Admin defines lifecycle policies. Organization Admin manages archival actions. Instructors retire outdated content. Learners access only active content. Support users assist archival recovery.

EPIC 9 — AI Personalization, Guidance & Intelligence

FEATURE 1: AI Buddy / Learning Assistant

Description

This feature provides an always-available AI assistant to guide learners through the platform.

It answers questions, explains concepts, and helps users navigate learning tasks.

It supports learning.

It does not replace instructors or grading.

User Journey

1. Learner opens the AI assistant.
2. Learner asks a question or requests help.
3. Assistant analyzes context and content.
4. Response is generated and shown.
5. Interaction is logged for improvement.

Exit state: Learner receives guided assistance.

User Map

Platform Admin defines assistant behavior and limits. Organization Admin enables or restricts assistant usage. Instructors guide how learners use the assistant. Learners interact directly with the AI Buddy. Support users review assistant-related issues.

FEATURE 2: Personalized Learning Paths

Description

This feature creates individualized learning paths based on learner goals, progress, and performance.

It replaces one-size-fits-all learning with adaptive sequencing.

User Journey

1. Learner profile and goals are analyzed.
2. Available courses and prerequisites are evaluated.
3. Personalized path is generated.
4. Learner follows recommended sequence.

Exit state: Personalized learning journey active.

User Map

Platform Admin defines personalization rules. Organization Admin enables personalized paths. Instructors align courses to paths. Learners follow recommended paths. Support users assist personalization issues.

FEATURE 3: Next-Best Lesson / Course Recommendations

Description

This feature recommends the most relevant next lesson or course for a learner.

It is driven by progress, performance, and behavior signals.

User Journey

1. Learner completes a lesson or activity.
2. System evaluates learner state.
3. Recommendation is generated.
4. Learner chooses to follow or ignore it.

Exit state: Context-aware recommendation delivered.

User Map

Platform Admin defines recommendation logic. Organization Admin reviews recommendation impact. Instructors influence recommendation inputs. Learners receive and act on recommendations. Support users assist recommendation issues.

FEATURE 4: Adaptive Content Suggestions

Description

This feature dynamically suggests content based on learner difficulty level and engagement.

It adjusts learning material in real time.

User Journey

1. Learner interacts with content.
2. Difficulty or engagement signal is detected.
3. Alternative or supplementary content is suggested.
4. Learner continues learning.

Exit state: Adaptive content delivered.

User Map

Platform Admin defines adaptation rules. Organization Admin enables adaptive learning. Instructors curate adaptive content. Learners receive suggestions. Support users assist content mismatch issues.

FEATURE 5: Skill Gap Analysis

Description

This feature identifies gaps between required skills and current learner proficiency.

It drives targeted learning recommendations.

User Journey

1. Learner skill data is evaluated.
2. Required skill benchmarks are compared.
3. Skill gaps are identified.
4. Learning actions are suggested.

Exit state: Skill gaps clearly identified.

User Map

Platform Admin defines skill models. Organization Admin maps required skills. Instructors align assessments to skills. Learners view skill gaps. Support users assist interpretation.

FEATURE 6: Automated Content Summarization

Description

This feature generates concise summaries of learning content using AI.

It supports faster revision and comprehension.

User Journey

1. Learner opens content.
2. Summarization request is triggered.
3. AI generates summary.
4. Summary is displayed.

Exit state: Content summarized for learner.

User Map

Platform Admin defines summarization limits. Organization Admin enables summarization. Instructors review summaries if needed. Learners use summaries for revision. Support users assist summary accuracy issues.

FEATURE 7: Smart Search & Q&A Over Content

Description

This feature enables AI-powered search and question answering across learning content.

It goes beyond keyword search.

User Journey

1. Learner enters a question.
2. AI searches relevant content.
3. Answer is generated with references.
4. Learner reviews result.

Exit state: Intelligent answer delivered.

User Map

Platform Admin configures AI search scope. Organization Admin enables smart search. Instructors ensure content quality. Learners ask questions and explore answers. Support users troubleshoot incorrect responses.

FEATURE 8: Engagement & Drop-off Prediction

Description

This feature predicts learner disengagement or drop-off risk.

It enables early intervention.

User Journey

1. Learner activity is monitored.
2. Engagement signals are analyzed.
3. Drop-off risk is predicted.
4. Alerts or actions are triggered.

Exit state: Risk of disengagement identified.

User Map

Platform Admin defines prediction thresholds. Organization Admin monitors engagement risk. Instructors intervene when alerted. Learners are passively analyzed. Support users assist false-positive handling.

FEATURE 9: Burnout & Overload Detection

Description

This feature detects signs of learner overload or burnout.

It balances learning intensity and well-being.

User Journey

1. Learning pace and load are tracked.
2. Stress or overload patterns are detected.
3. Alerts or adjustments are suggested.
4. Learning intensity is moderated.

Exit state: Burnout risk mitigated.

User Map

Platform Admin defines burnout indicators. Organization Admin monitors learner well-being signals. Instructors adjust workload. Learners receive recommendations. Support users assist escalation.

FEATURE 10: AI Nudges & Reminders

Description

This feature sends intelligent nudges to encourage learner action.

It replaces generic reminders with context-aware prompts.

User Journey

1. Trigger condition occurs.
2. AI selects appropriate nudge.
3. Message is delivered.
4. Learner responds or ignores.

Exit state: Timely learner engagement prompt delivered.

User Map

Platform Admin defines nudge policies. Organization Admin customizes nudges. Instructors influence trigger conditions. Learners receive nudges. Support users assist notification issues.

FEATURE 11: Learning Outcome Insights

Description

This feature analyzes learning outcomes using AI.

It identifies what learning approaches work best.

User Journey

1. Learning and assessment data is collected.
2. AI analyzes outcome patterns.
3. Insights are generated.
4. Insights are presented to stakeholders.

Exit state: Actionable outcome insights available.

User Map

Platform Admin accesses global insights. Organization Admin views organizational outcomes. Instructors use insights to improve teaching. Learners do not directly access insights. Support users assist reporting issues.

FEATURE 12: Explainable Recommendations

Description

This feature explains why AI recommendations are made.

It builds trust and transparency.

User Journey

1. Recommendation is shown.
2. Learner requests explanation.
3. AI explains reasoning.
4. Learner decides action.

Exit state: Recommendation understood.

User Map

Platform Admin enforces explainability standards. Organization Admin ensures transparency compliance. Instructors reference explanations. Learners view explanations. Support users assist trust issues.

FEATURE 13: AI Model Feedback & Tuning Loop

Description

This feature collects feedback to improve AI models over time.

It enables continuous learning and refinement.

User Journey

1. AI interaction occurs.
2. Feedback is collected explicitly or implicitly.
3. Data is analyzed.
4. Models are tuned and updated.

Exit state: AI models improved.

User Map

Platform Admin oversees AI tuning and governance. Organization Admin provides feedback context. Instructors contribute qualitative feedback. Learners provide usage signals. Support users assist model issue reporting.